

Log Scale Surprise Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Medium	ID	log-scale-surprise

Big idea: Mathematical representations can make the same relationship look explosive, steady, or manageable.

Student-Facing Prompt

Inspect Log Scale Surprise Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Log Scale Surprise Stimulus A	Student-facing stimulus 1: Dataset: 1, 2, 4, 8, 16, 32, 64 over seven days. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Log Scale Surprise Stimulus B	Student-facing stimulus 2: Question: does the graph show runaway growth, steady multiplicative growth, or both?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Log Scale Surprise Reveal / Twist	Student-facing stimulus 3: Notation card: $\log_{10}(1000) = 3$ shows how a multiplicative change becomes an equal step on a log scale. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Mathematical representations can make the same relationship look explosive, steady, or manageable.

Actual Lesson Questions

#	Question for students
1	After inspecting Log Scale Surprise Stimulus A, what is your first knowledge claim?
2	Which exact detail from Log Scale Surprise Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	How does mathematical representation shape understanding?

Ready-to-Use Examples

- Dataset: 1, 2, 4, 8, 16, 32, 64 over seven days.
- Question: does the graph show runaway growth, steady multiplicative growth, or both?
- Notation card: $\log_{10}(1000) = 3$ shows how a multiplicative change becomes an equal step on a log scale.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

1. Give students a simple growing dataset.
2. Groups plot or inspect the same data on a linear graph and a logarithmic graph.
3. Students describe the story each graph seems to tell.
4. Groups identify which graph is better for different purposes.
5. Debrief how mathematical representation affects knowledge claims.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: How does mathematical representation shape understanding?
- Can one graph be more truthful than another if both are accurate?
- What responsibilities come with choosing a scale?

Knowledge Questions

- How does mathematical representation shape understanding?
- Can accurate graphs tell different stories?
- What makes a graph responsible?

Transfer Example

For example, run this activity with the same growth data plotted on linear and logarithmic scales. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...